

WHAT I LEARNED FROM ASSIGNMENT 6 – SHOPLY

Flutter E-Commerce Application

1. DATA MODELS AND APPLICATION DATA

I learned how to create a model class to represent real-world data. For Shoply, I created a Product model containing information such as product name, description, price, category, image, rating, reviews, featured status, and new-product status.

Instead of writing product information directly inside the UI, I stored the products separately in a product data file. This makes the application cleaner and easier to maintain.

2. SEARCH, FILTERING AND SORTING

One of the main things I learned was how to make product catalogs interactive.

The Shoply catalog allows users to search for products by name or description. I also implemented category filtering so users can view products from categories such as Electronics, Fashion, Books, and Accessories.

I also learned how sorting can be implemented to arrange products based on different conditions such as price or rating. This helped me understand how the same list of objects can be processed dynamically according to user actions.

3. STATE MANAGEMENT USING SETSTATE

I learned how Flutter's `setState()` can be used to update the interface when application data changes.

For example, when a user adds a product to the wishlist or cart, the UI automatically updates. The same concept is used for search queries, category selection, sorting, and other interactive features.

This helped me understand the basic concept of state management in Flutter.

4. SHOPPING CART AND WISHLIST

- I learned how to implement common e-commerce features such as a wishlist and shopping cart.

- The wishlist allows users to save products they are interested in.
- The cart stores products selected for purchase.
- I also learned how to calculate values such as subtotal, delivery charges, and total price dynamically from the cart data.

5. NAVIGATION BETWEEN SCREENS

The application contains different screens for different parts of the shopping experience.

I learned how Flutter navigation can be used to move between screens, such as the catalog and cart.

This showed me how a larger application can be divided into multiple screens instead of keeping everything inside one file.

6. RESPONSIVE UI DESIGN

Another important learning from this assignment was responsive design.

I used Flutter's layout tools and screen width information to change the interface according to the available screen size.

The product grid changes the number of columns depending on the screen width. The navigation and other sections also adapt for smaller devices.

This helped me understand how to create applications that work properly on desktop, tablet, and mobile screens.

7. CUSTOMSCROLLVIEW, SLIVERS AND GRID LAYOUT

I learned about CustomScrollView and Sliver widgets for building a scrollable and structured catalog page.

I also used GridView/SliverGrid to display products in a professional e-commerce layout.

Using these widgets helped me understand how Flutter can efficiently display large collections of items while maintaining a responsive design.

8. HERO CAROUSEL AND ANIMATIONS

I implemented a featured-product carousel using PageView.

The carousel automatically changes featured products and also provides navigation controls and indicators.

Through this, I learned about PageController, timers, animations, and the importance of disposing controllers and timers properly.

9. REUSABLE WIDGETS AND PROJECT STRUCTURE

I learned why reusable widgets are important in Flutter.

Instead of putting the complete application inside main.dart, I separated the project into models, data, screens, widgets, and utility files.

For example, the hero carousel and footer are separate reusable widgets.

This makes the project easier to understand, debug, modify, and expand.

10. THEMING AND UI DESIGN

I learned how to create a consistent design system using a central theme.

Shoply uses its own color palette, typography, backgrounds, buttons, cards, borders, and other UI elements.

Keeping these design values organized makes the application look more professional and prevents inconsistent styling throughout the project.

11. USER FEEDBACK AND FORM VALIDATION

I also learned how applications should provide feedback when users perform actions.

Shoply uses SnackBars, dialogs, and visual changes to inform users about different actions.

I also implemented newsletter input validation, which helped me understand how user input can be checked before processing it.

12. OVERALL LEARNING

The biggest learning from this assignment was understanding how multiple Flutter concepts work together to create a complete application.

Instead of only creating a static UI, I worked with real application logic such as searching, filtering, sorting, wishlist management, shopping cart functionality, navigation, responsive layouts, and user interaction.

This assignment improved my understanding of Flutter development and

helped me move from creating simple screens toward building applications with a more realistic structure and user experience.

CONCLUSION

Assignment 6 gave me practical experience in developing an e-commerce application using Flutter. I learned how to organize application code, manage data and state, create reusable components, build responsive layouts, and implement interactive features.

Overall, Shoply helped me understand how the concepts learned in Flutter can be combined to build a complete and professional-looking application.